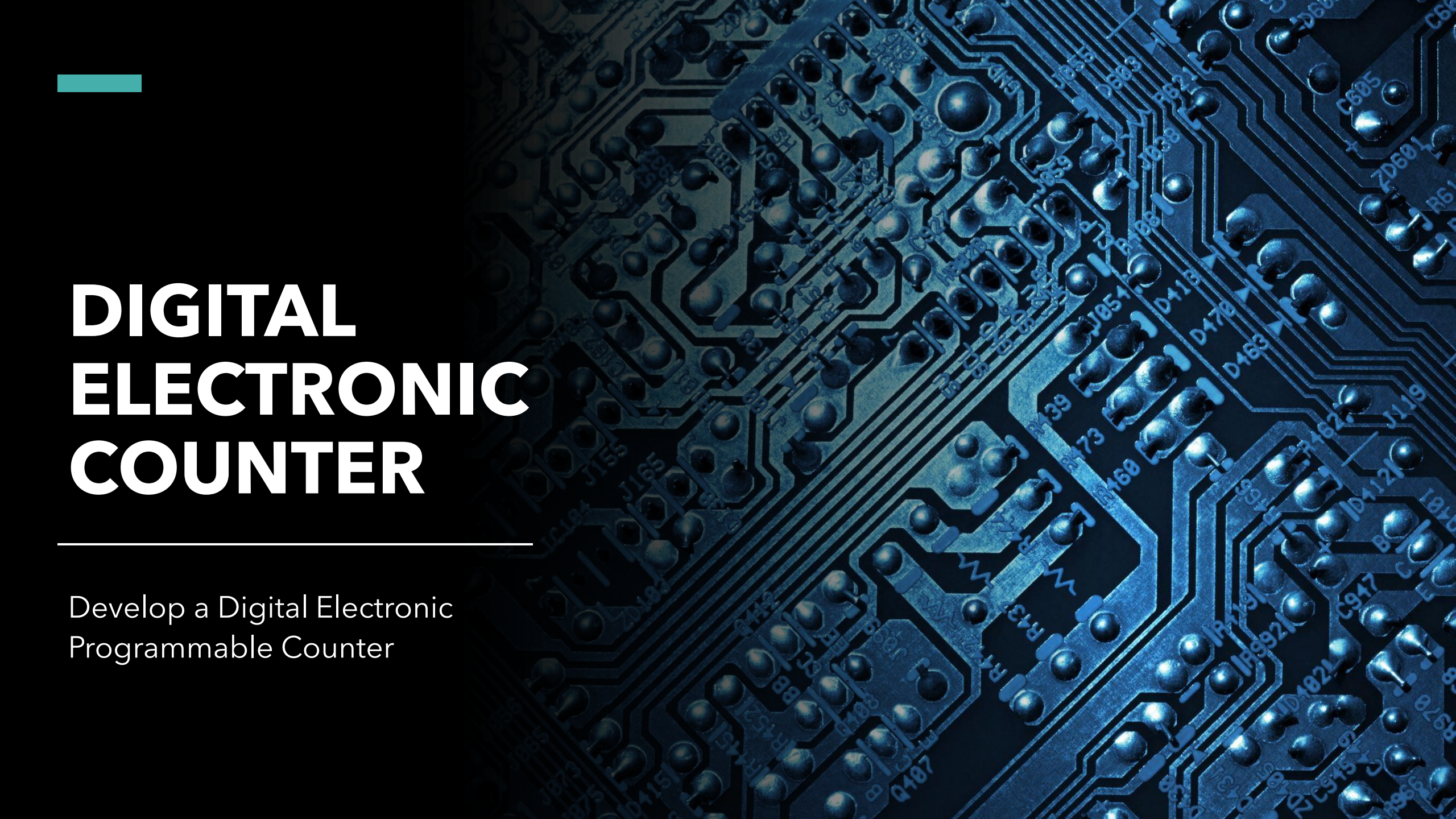
DIGITAL ELECTRONIC COUNTER

Develop a Digital Electronic Programmable Counter



SCOPE

Scope of Device

Develop an electronic digital counter based device that can perform the following functions:

1. Program up to 10 pre-programmed functions for counting against time or clicks on hardware buttons
2. Allow Timers (upto two) to Count up or down based on the user input and or run simultaneously
3. Allow Hardware based buttons to count the number for each programmed function
4. Store upto 30 previous memories of the counted functions and rounds
5. Allow changes and updated to be made based on firmware updates
6. Allow integration of hardware button and touchscreen to control and program the device.

Scope of Work: (Detailed on following pages)

1. Select Components
2. Develop Schematic Design
3. Design Layout
4. Firmware/Software Development
5. Final integration Testing



Clear Image and backdrop data / information will be shared on award of contract.

Reference Image (Page 8)

1. Select Components

1. Carry out analysis and feasibility of electronic components to choose those that best fit requirements. Requirements will be shared in detail once the contract is established on upwork.
2. Provide a comparison of cost of components to select most efficient but cost effective components.
3. Checking and assure availability of selected components / modules in order to avoid inconvenience later.
4. Submit detailed document that outlines the component, manufacturer, cost and links to procduct data/sspec sheet.

2. Develop Schematic Design

1. Designing circuit for implementing main application and hardware requirements.
2. Developing library (if needed) for selected components to create a schematic.
3. Selecting loss free and power efficient approach in circuit design.
4. Calculating power requirements of the circuit in order to not provide extra power or less power in terms of current consumption.
5. Finalizing and reviewing the circuit before designing PCB.

3. Design Layout

1. Designing a Printed Circuit board based on the created schematic.
2. Ensure proper and effective placement of components keeping in mind the requirement of main application and hardware as well as electronic recommendations.
3. Laying out proper tracks for current flow based on amount of current flow through the PCB as needed.
4. Finalizing and review the PCB design, fit form and proposed enclosure shell for manufacturing.

4. Firmware/Software Development

1. Write firmware according the selected layout and in-line with the requirements of the product. Client will share programming guide and matrix for what the programming will include along with programming parameters for each domain.
2. Compile and remove any logical or syntax errors in the firmware.
3. Burn the firmware onto the hardware and then test whether it works according to the functionality.

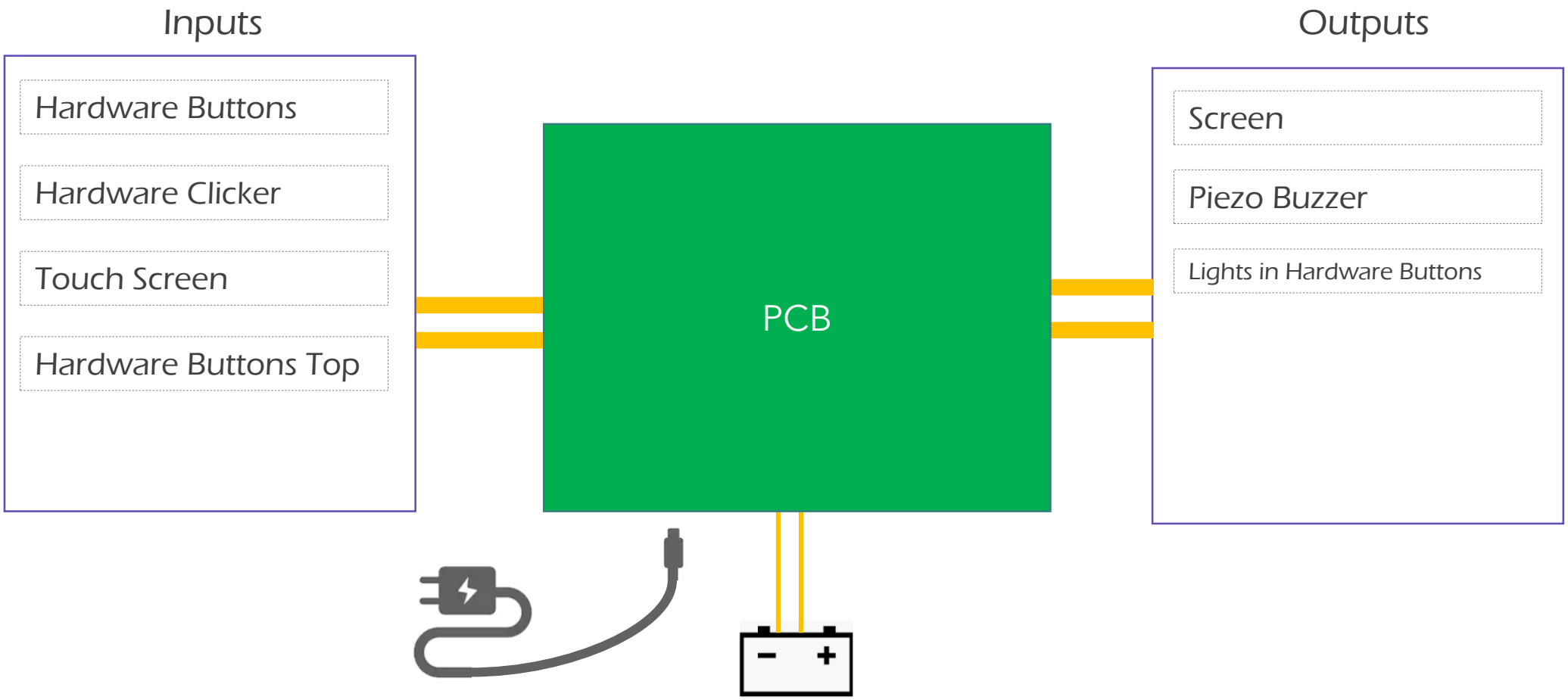
5. Final integration Testing

1. Complete testing of the PCB, ensuring all the functionality is according to the requirement.
2. Eliminate any issues that maybe encountered during testing and debugging.
3. Finalize and install the Hardware for usage.
4. Combined testing of the hardware and the software and ensuring all requirements are met.
5. Providing timely updates for all the ongoing activity.

Deliverables

1. Parts File
2. Libraries
3. Circuit Design File
4. Layout File
5. Gerber Files
6. NC Drill File
7. Pick & Place File
8. Bill of Materials
9. STEP File
10. Source Code Files
11. Weekly Status Updates

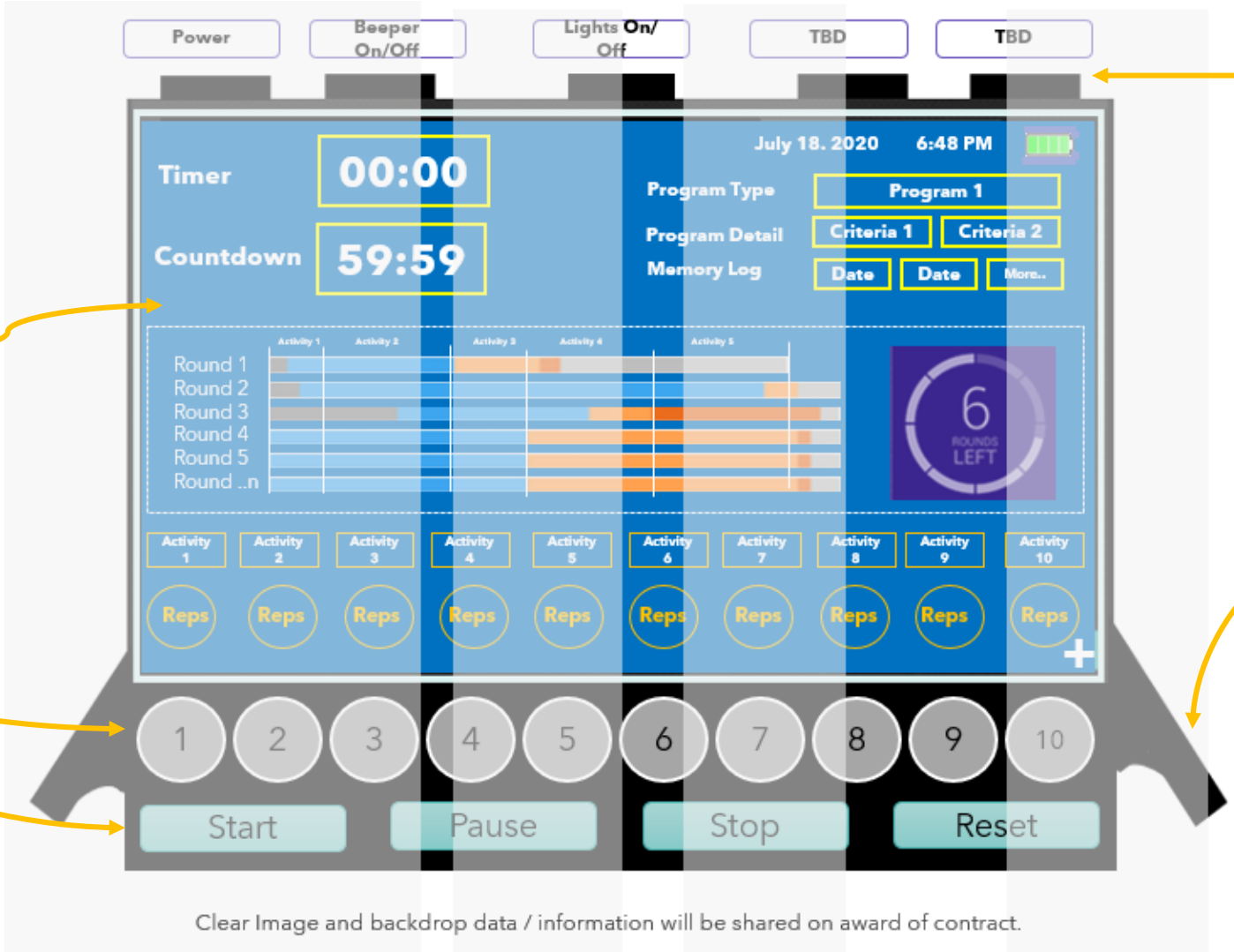
Inputs / Outputs



Concept Layout

Touch Screen
(1024 x 600)

Hardware
Buttons



Hardware
Buttons Top

Hardware
Clicker

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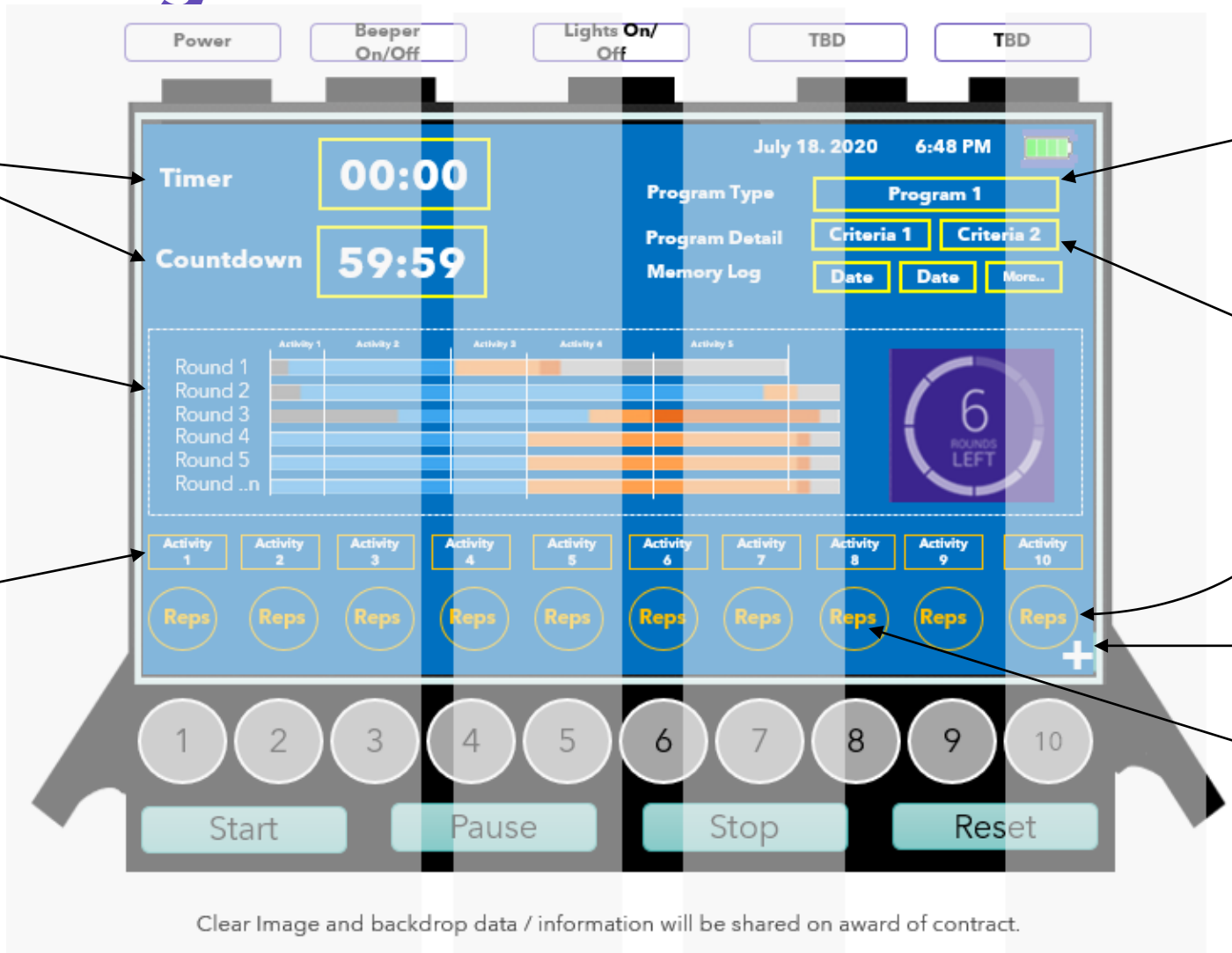
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Required Program / Function

Regular Stop Watch Timer

This is the output screen which will log the time for when each activity is completed.

There are about 150 pre-programmed activities that will be available for user to choose and they will be able to select a maximum of 10 on current screen.



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Program Type will be based on predetermined programs that will be around 50 in number and a document listing all programs will be provided

Program detail further outlines the detail which adds / subtracts / multiples or divides the reps

Plus sign will be able to add additional ten items for next … up to a total of 30 total activities.

These are the reps which will be the numbers which can be entered using on screen keyboard or can be scrolled range 1- 2000

1. User Selects Program Type

2. User Selects Program Detail (Criteria) which is a numerical number based on the program.

3. User Selects activity from a predefined list of activities.

4. User selects reps which is a numerical number range from 1 - 2000

5. User programs next activities from 1 (min) to 30 (max)

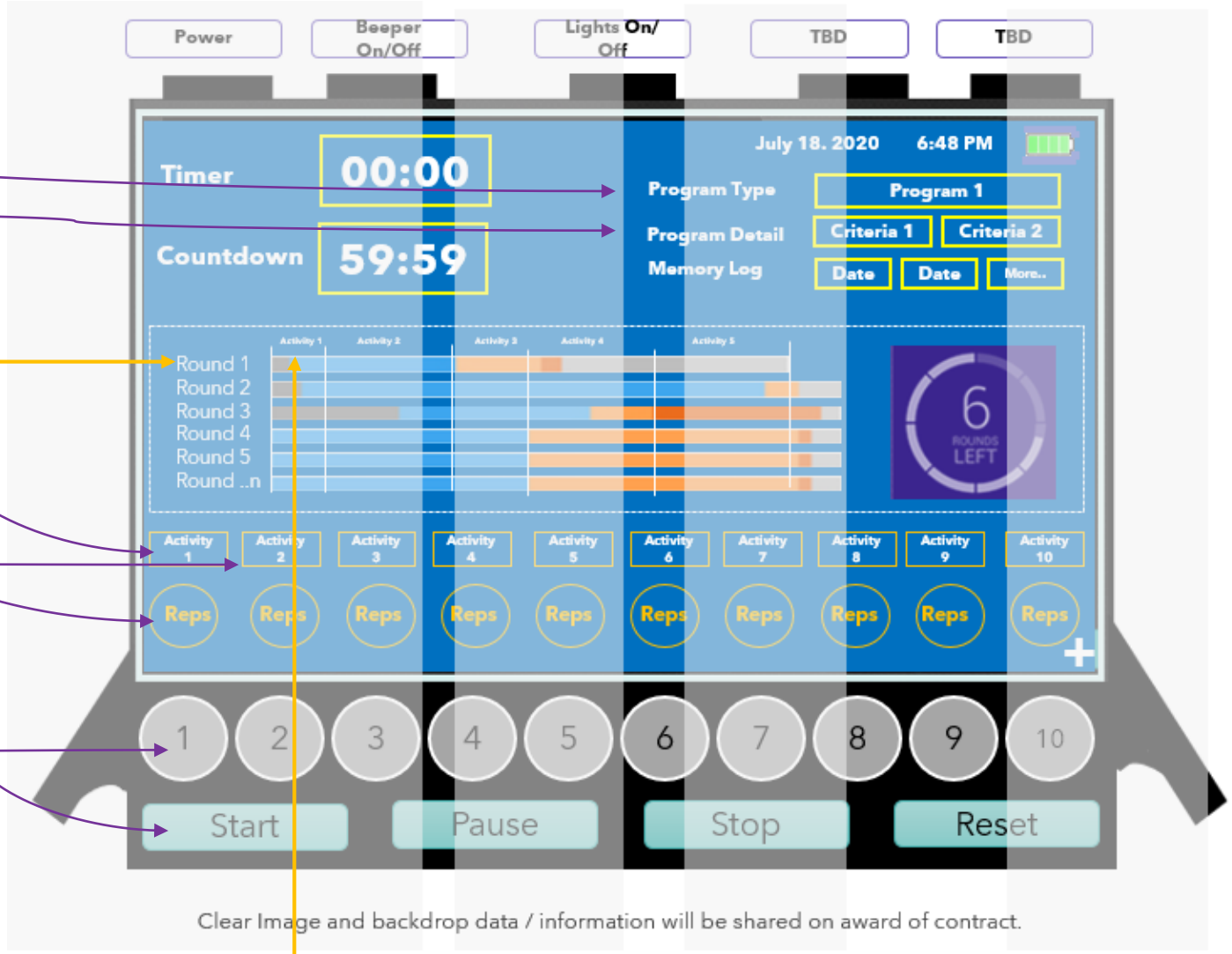
6. User hits start button and the timer / countdown starts.

7. User hits hardware button under each activity to record the time.

8. Once all buttons for programmed activity is pressed it recorded as a round.

9. Each logged activity time is shown on the round screen.

8. This is a mix of programmable calculator and stop watch that records activities.



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User Example

This example shows a user has programmed four activities which are :

1. Drive 20 Meters
2. Walk 6 Meters
3. Sit 10 Minutes
4. Jump for 15 Seconds

Reps ~~for each activity is programmed~~ under it.

As each activity is completed – the hardware button is pressed and it records the time and once all activities are completed it shows the total round and time it took for each activity in the round.



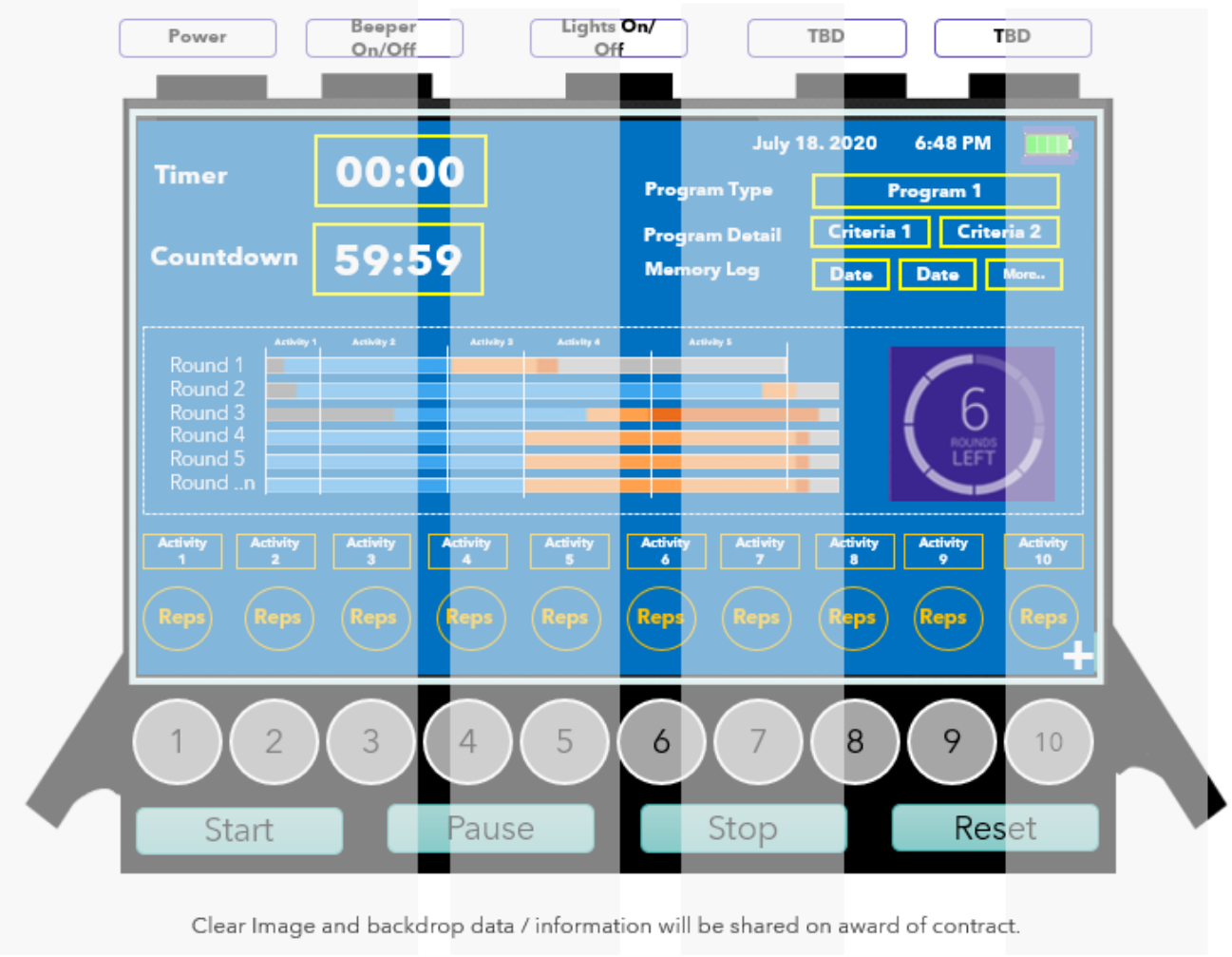
Customizable Interface

1. Plan for 3 different pages/layout for the screen interface.

2. Plan for the memory function to take a screenshot of the screen.

3. The background can be a wallpaper that the user can change by uploading pictures via usb-c cable.

4. 4. The user should be able to use on screen keyboard to type in custom program or activity name and should be able to enter the numbers using on-screen numerical keypad.



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